

NIA Challenge Phase 3

SpeechCARE Team

Columbia University Irving Medical Center, New York, NY 10032, United States

Maryam Zolnoori, PhD. Assistant Professor, Speech processing Lab, email:mz2825@cumc.columbia.edu

Hossein Azadmaleki, CS. Project management, Development of SpeechCARE models, Contributed to: Multi-task learning, Continual learning, SpeechCARE Robustness, and Report drafting, - h.azadmaleki@gmail.com

Yasaman Haghighin, CS. Preprocessing components, Multi-task learning, Trustworthy, and Generatability.

Fatemeh Taherinezhad CS. Multi-task learning, Continual learning, and Trustworthy.

Maryam Dadkhah CS. Explainability framework, figure designs

Ali Zolnour, CS. Model generalizability on Home Care speech data.

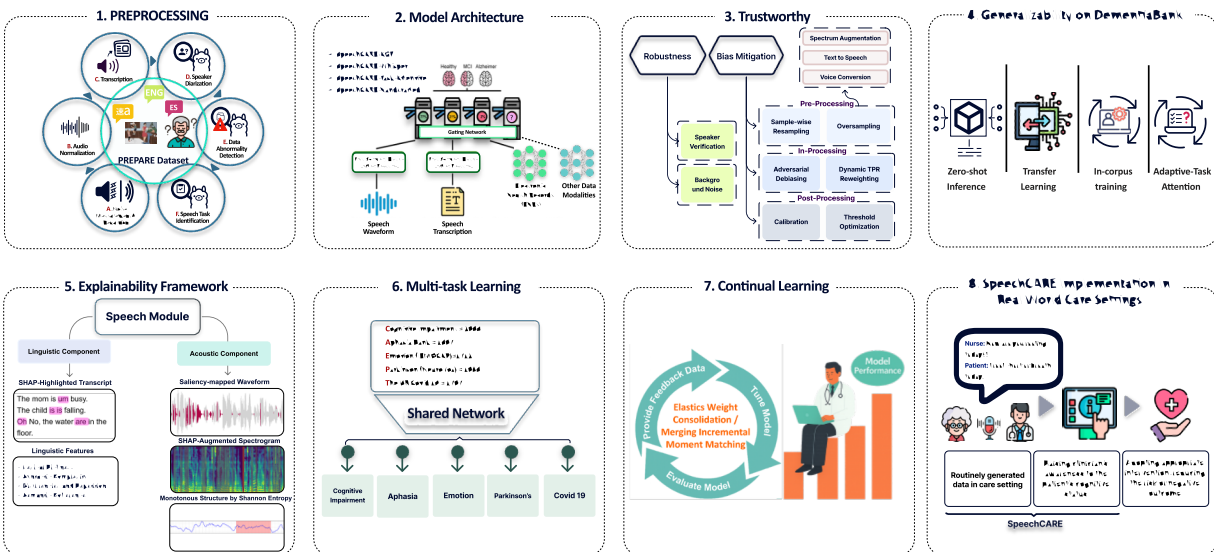
Sina Rashidi, CS. Preprocessing components and Generatability.

Mohammad Javad Momeni Nezhad, CS. Continual Learning

* CS: Computer Science

Overview:

In the NIA Challenge Phase 3, SpeechCARE presents a strong speech processing pipeline that introduces innovations across eight key areas: (1) SpeechCARE Preprocessing ([notebook: preprocessing_scripts.ipynb](#)), (2) SpeechCARE Models ([notebook: SpeechCARE_script.ipynb](#)), (3) SpeechCARE Trustworthy ([notebook: Trustworthiness_script.ipynb](#)), (4) SpeechCARE Generatability ([notebook: Generalizability_script.ipynb](#)), (5) SpeechCARE Explainability ([explainability.md](#)), (6) SpeechCARE Multi-task Learning ([notebook: multi_task_learning_script.ipynb](#)), (7) SpeechCARE Continual Learning ([notebook: continual_learning_script.ipynb](#)), and (8) SpeechCARE-Application in real-world care settings ([notebook: Homecare_script.ipynb](#)). Figure below illustrates an overview of our framework.



1) SpeechCARE Preprocessing Component. Real-world clinical recordings often contain background noise, additional speakers, along with multiple speech tasks. To manage these variabilities, SpeechCARE’s preprocessing includes: (a) Noise Measurement & Reduction: A low-pass¹ filter removes noise above 8 kHz, preserving critical prosodic features. (b) Peak normalization²: Audio samples are scaled to their maximum amplitude, reducing variabilities caused by recording devices. (c) Transcription: Audio are transcribed using Whisper-large-v3³. With the peak normalization step, Word Error Rate⁴ (WER) dropped from 0.141 to 0.137 (English), 0.434 to 0.356 (Spanish), and 2.80 to 0.860 (Mandarin) compared to Phase 2 results. (d) Speaker Diarization: A two-step pipeline uses WhisperX⁵ for word-level timestamps, followed by Llama-3.1-405B⁶ to diarize and tag each sentence as patient or clinician. On Pitt corpus⁷, it surpassed the SOTA model, NVIDIA NeMo⁸ (WDER 0.09 vs 0.15; SpkCnt-MAE 0.10 vs 0.73). (e) Data Abnormality Detection: LLaMA-3.3-70B⁹ with prompt engineering and few-shot learning is used to detect corrupted samples (e.g., dropped audio), achieving 100% precision on the Phase-2 dataset. (f) Speech Task Identification: LLaMA-3.3-70B, with prompt engineering, identifies speech tasks within long interviews and marks the start and end points of each task. This achieved 100% accuracy on the Delaware corpus¹⁰ (**Table 1**). **Performance and Innovations.** Preprocessing enhancements in Phase 3 increased SpeechCARE’s F1-score on the Phase-2 dataset from 0.72 to **0.73** (with SpeechCARE-AGF). The addition of LLaMA modules (d-f) enabled preprocessing of new speech datasets, like the Delaware.

2) SpeechCARE Models. SpeechCARE primarily consists of: (1) **A Feature network:** Linguistic cues are extracted with transformer models. Acoustic cues are encoded with speech transformer models including mHuBERT¹¹ (accepts waveforms) or Whisper (accepts mel-spectrograms). To use speech transformers effectively, audio is segmented (5s for mHuBERT, 30s for Whisper) to match pretraining spans so each window remains within the model’s pretraining range, avoiding positional drift; a learnable [CLS] token is prepended, and a lightweight self-attention block generates an utterance-level embedding. Demographic and clinical features are also encoded by MLP networks and treated as additional modalities. **Fusion Network:** The Adaptive Gating Fusion (AGF) employs a soft attention module¹² as a gating network and dynamically prioritizes acoustic, linguistic, and clinical cues based on their diagnostic value and quality. Four implementations follow this design: (1) **SpeechCARE-AGF**, which combines mHuBERT, mGTE¹³, and demographic vectors; (2) **SpeechCARE-Whisper**, in which, we omit linguistic transformers because the AGF module consistently down-weighted transcript embeddings; (3) **Novel SpeechCARE-Task-Attentive**, introduced in Phase 3, targets corpora with multiple speech tasks per participant (e.g., Delaware). First, all task-specific transcripts are concurrently encoded using a lightweight linguistic transformer and then a Mixture-of-Experts¹⁴ gating network weights the resulting embeddings. The weighted sum of these embeddings forms a unified representation which is then fused with acoustic embeddings (through AGF) and finally classified by an MLP. This lightweight architecture also provide interpretable task-importance weights that can guide future data-collection design. (4) **SpeechCARE-Handcrafted**, designed for low-resource settings, uses 5,967 handcrafted acoustic features, capturing diverse vocal attributes. It eliminates the need for large neural models, offering greater transparency for clinical interpretation.

3) SpeechCARE Trustworthy. In Phase 3, SpeechCARE integrates the SpeechCURA toolkit—designed to mitigate demographic bias. This is achieved through a three-stage process: (1) **Preprocessing**, which expands underrepresented groups via (i) spectrogram augmentation¹⁵ (frequency/time masking and shifting), (ii) LLM-TTS speech synthesis (LLaMA-3-8B extends transcripts and Spark-TTS¹⁶ converts them to speaker-matched speech), and (iii) voice conversion (FreeVC¹⁷ transforms the source speaker’s voice to resemble a target speaker). (2) **In-processing**, using (i) an adversarial network¹⁸ to suppress irrelevant attributes (like gender); and (ii) dynamic reweighting¹⁹ to upweight loss for underrepresented subgroups. (3) **Post-processing**, employing (i) Isotonic Calibration²⁰ and (ii) group-specific Threshold Optimization²¹ to balance sensitivity and specificity. Additionally, to improve robustness to background noise, we injected real-world noise (e.g., conversation; see Fig. 1) into the Phase-2 dataset to create a noisy test set. We then assessed model performance when trained on the original training data and when trained on noise-augmented training data. The latter was intended to increase robustness to noisy audio. We also

embedded a speaker verification component, enabling SpeechCARE to perform subject attribution for longitudinal tracking and other use cases. This module extracts speaker embeddings using Pyannote-audio²² and links test utterances to known subjects when cosine distance ≤ 0.33 . **Performance and Innovations. (a) Bias Mitigation:** In the Phase-2 dataset, voice conversion achieved the greatest bias reduction (true-positive-rate-gap dropped 2% on average, 7% max on age group), but other methods only yielded marginal improvements. This result is driven by our Speaker-Pair-Selection method: each speaker is encoded as a vector of shimmer, mean MFCCs, and speech rate. Cosine distances are computed between all pairs, and maximum-weight matching selects disjoint pairs with maximal acoustic dissimilarity. Training on these high-contrast pairs outperforms random speaker pairing. **(b) Robustness–Noise Tolerance:** On the Phase-2 dataset, SpeechCARE-AGF outperformed SpeechCARE-Whisper overall across different Signal-to-Noise Ratio levels (**Fig. 1**), largely due to its AGF fusion and gating network, which down-weights noisy acoustic inputs in favor of stable transcript inputs. Noise-augmented fine-tuning with +5k synthetic clips at SNR 5, further improved F1 by 5.5% for SpeechCARE-AGF and 3.5% for SpeechCARE-Whisper. **(c) Speaker Verification:** On the Delaware dataset (**Table 1**), we observed 100% verification accuracy, supporting secure longitudinal monitoring.

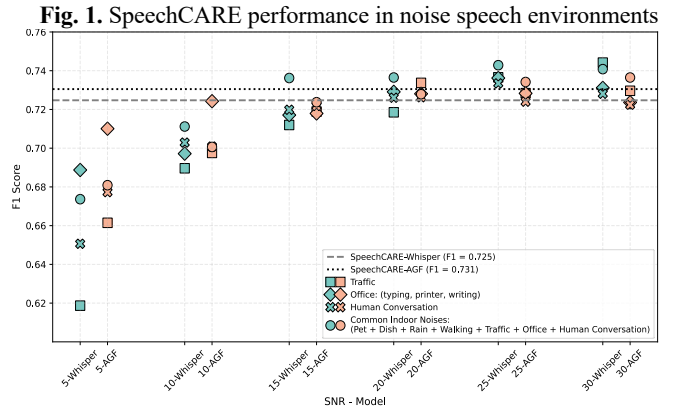


Table 1. Dataset Characteristics and Generalizability Results of SpeechCARE on DementiaBank Corpora

Corpus	Participants	Language/Speech Tasks	Overlap with Phase-2 Dataset	Strategy of Evaluation	Best Performing Model	Performance
WLS	1,368 HC	English/Cookie-Theft Picture	832; 538 reserved	Zero-shot Inference	AGF	F1= 0.99
Lu	51 ADRD	Mandarin/Cookie-Theft Picture	None	Zero-shot Inference	AGF	F1= 0.98
PerLA	21 ADRD	Spanish/Personal Narrative	None	Zero-shot Inference	AGF	F1= 0.96
Chou	40 HC, 47 MCI	Mandarin/ Picture description: Daddy, Park, Night-market	None	Zero-shot Inference	AGF	F1= 0.09
				Transfer learning	AGF	F1= 0.84
				In-corpus training	AGF	F1= 0.92
				Task-Attentive	AGF	F1=0.82
Delaware	106 HC, 99 MCI*	English/ Picture description: Cookie Theft, Cat Rescue, Rockwell; Story recall: Cinderella, Procedural Narrative: PB&J Sandwich	81 participants; 113 recordings (some recorded two sessions)	Zero-shot Inference	AGF	F1=0.41
				Transfer learning	AGF	F1=0.70
				In-corpus training	AGF	F1= 0.78
				Task-Attentive	AGF	F1=0.78

* A pair of duplicate IDs with inconsistent labels was removed from the data

4) SpeechCARE Generalizability on DementiaBank. SpeechCARE was evaluated on five DementiaBank corpora to assess generalizability (**Table 1**). Single-class corpora (WLS²³, Lu²⁴, and PerLA²⁵) were only used for inference. In corpora with control and MCI samples (Delaware and Chou), test sets (20%) were split in a way that avoids any speaker overlap with Phase-2 dataset. On these corpora, we evaluated SpeechCARE-Whisper and AGF using four strategies: **(1) Zero-shot inference**—applying the Phase-2-trained model without fine-tuning; **(2) Transfer learning**—fine-tuning the Phase-2-trained model on the target corpus; **(3) In-corpus training**—skipping Phase-2 data, the model is fine-tuned solely on Delaware and Chou, assessing the benefits of longer audios (>30s) for each speech task; participant-level results were obtained via majority voting across tasks; **(4) Task-Attentive**, (see SpeechCARE Models) for Delaware and Chou with multiple tasks per participants. **Performance and Innovations.** See Table-1 for results. Findings highlight SpeechCARE’s generalizability across languages, tasks, and recording conditions—supporting its utility in real-world, multilingual clinical settings.

5) SpeechCARE Explainability Framework. For Phase 3, we conducted usability interviews with clinicians at Columbia Medical Center (IRB #AAAV8664) that helped refine the explainability framework introduced in Phase 2. Specifically, we implemented: **(1) Summary of important findings,**

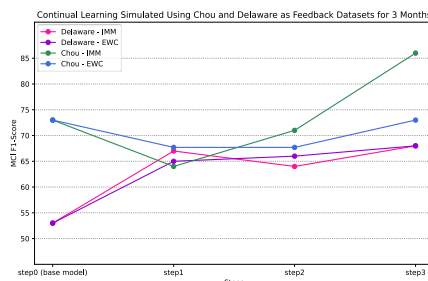
including (a) model outcome and confidence score; (b) the four most influential clinical and speech risk factors; (c) the relative contribution of each data modality to the model’s decision; **(2) Expandable interpretation modules**, including (a) *Clinical Biomarkers*, identify Important Electronic Health Records (EHR)²⁶-derived clues (e.g., sleep disruption, social isolation) using LLaMA-3.3-70B, trained with expert prompting and Retrieval-Augmented Generation²⁷ over structured EHR data; (b) *Linguistic Biomarkers*, combines SHAP²⁸ token-level attribution and LLaMA-3.3-70B (trained with Tree-of-Thought²⁹ reasoning) to highlight key linguistic cues (lexical diversity, fluency, syntactic structure, and semantic coherence); (c) *Acoustic Biomarkers*, integrate mHuBERT gradient saliency maps³⁰ to highlight informative waveform segments, and SHAP-annotated spectrograms to visualize frequency-energy patterns, along with plotting fundamental frequency and informative pauses (e.g., pre-noun pauses). Rhythmic structure is analyzed using Shannon entropy to detect monotonous segments of voice. A prototype dashboard, along with codebooks, and hypothetical examples, was submitted with this report.

6) SpeechCARE Multi-Task Learning. Many acoustic and linguistic markers of ADRD overlap with other medical conditions (like Aphasia). Leveraging this overlap, Multi-Task Learning (MTL) offers a promising solution by enabling models to learn shared patterns across related conditions and improve generalizability. We implemented an MTL pipeline using speech samples from five datasets: Cognitive Impairment (phase-2 dataset, n=2,068) and using 5,961 samples from four auxiliary datasets: AphasiaBank³¹ (aphasia, n=1,507), IEMOCAP³² (Emotion, n=1,712), NeuroVoz³³ (Parkinson, n=1,035), and UK Covid-19 Audio Dataset³⁴ (n=1,707). The SpeechCARE-MTL built on two architectures (AGF and Whisper) and each included a shared feature network, task-specific output heads, an adversarial domain discriminator³⁵ (that promote task-invariant representations), and Homoscedastic Uncertainty³⁶ for loss balancing. **Performance and Innovations.** Whisper consistently outperformed AGF across all auxiliary tasks. ADRD F1 scores were highest with AphasiaBank (73.79), followed by IEMOCAP, NeuroVoz, and UK-Covid (0.7233). Auxiliary task F1-scores were 0.8349, 0.6187, 0.7561, and 0.6044, respectively. This study is the first application of MTL to ADRD screening that explicitly leverages clinically related speech patterns.

7) SpeechCARE Continual Learning. ADRD speech models, trained on curated corpora, confront new patient demographics and clinician feedback after deployment. In such situations Continual learning³⁷ (CL) can help models adjust to feedback while preserving prior knowledge. To evaluate this ability we used Elastic Weight Consolidation³⁸ (EWC) and Incremental Moment Matching³⁹ (IMM) CL algorithms. EWC uses the diagonal Fisher Information Matrix plus a quadratic penalty to limit changes to task-critical weights, integrating clinician labels without forgetting. IMM treats successive parameter sets as Gaussian posteriors and fuses them via mean-IMM or Fisher-weighted mode-IMM, providing lightweight Bayesian updates that curb catastrophic forgetting. **Performance and Innovations.** See Fig. 2 for results. We simulated two deployments: an English clinic with the Delaware dataset (n=205, Table 1) and a Mandarin clinic with the Chou dataset (n=87, Table 1). Over three months, feedback was simulated by randomly selecting one training subset per month; each subset triggered an incremental EWC or IMM update. Models were tested after every round on the same hold-out test sets used in the earlier generalizability study in section 4. IMM consistently outperformed EWC in both settings, though both trailed the transfer-learning baseline. Innovation. This study is the first to apply a continual learning method for speech processing algorithms for ADRD detection.

8) SpeechCARE- Application in Real-World Care Settings. Supported by NIH K99/R00 (AG076808) and A2 Collective (P30-AG-073105), SpeechCARE was deployed in home healthcare to detect mild cognitive impairment (MCI). Among 443 patients contacted, 135 enrolled; 113 completed in-home visits, 98 joined follow-up calls, and 47 participated in nurse-patient recordings. SpeechCARE achieved AUCs of 80.76, 81.13, and 86.47 across the three settings, respectively. This is the first study to evaluate cognitive-linguistic signals across naturalistic speech contexts, supporting passive, scalable MCI screening in real-world care workflows

Fig. 2. Continual learning results



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